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An Introduction to Natural Language Processing

NLP and Digital Traces



By **Sukanya Nath**

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About Me



Sukanya Nath

Education

2018-2021	Ph.D. in Computer Science (Computational Linguistics)	@ University of Neuchatel
2016-2018	MSc. In Computer Science (Data Science Specialization)	@ University of Bern
2009-2013	BTech in Computer Science	@ National Institute of Technology, Silchar, India

Teaching

Since 2023	Module Lecturer (CAS NLP)	@ University of Bern
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Work Experience

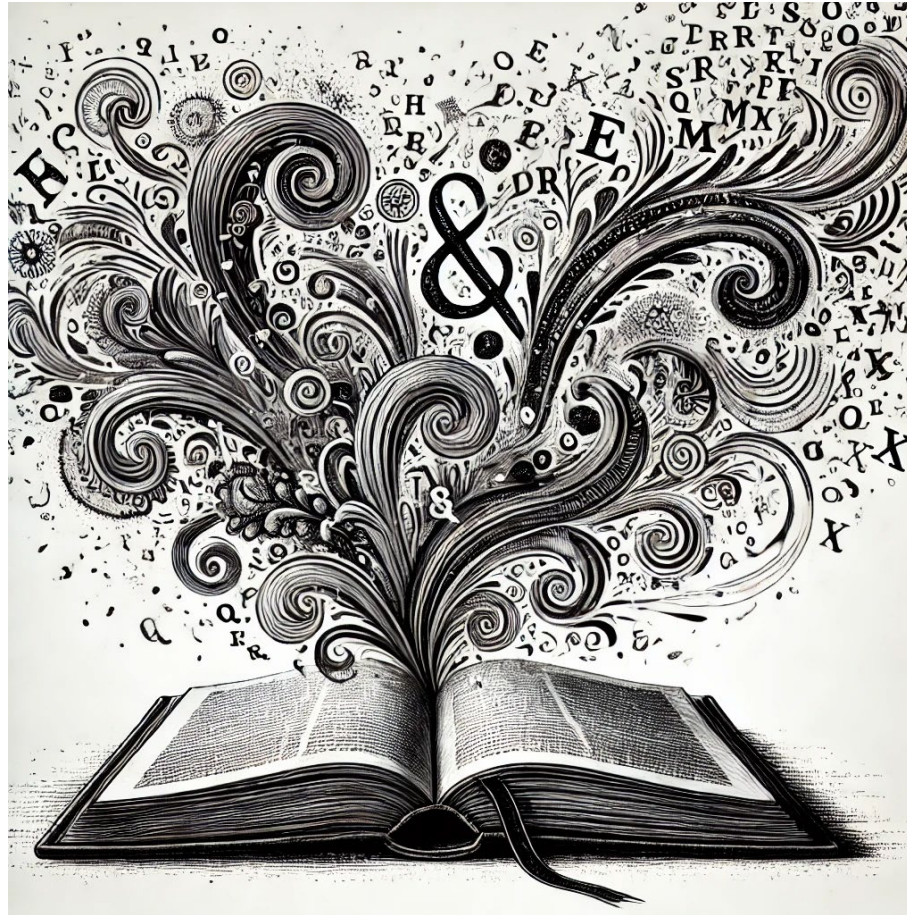
2023-present	Scientific Collaborator at Data Science Lab	@ University of Bern
2021-2023	Scientific Collaborator at IFeL	@ Swiss Distance University of Applied Sciences Brig
2020-2020	Research Assistant	@ University of Applied Sciences Bern
2016-2017	Data Science Intern	@ BEDAG AG
2013-2015	Software Developer	@ Samsung Research Institute India

Session Outline

- Part 1 (Introduction to NLP)
 - Language and a brief history of NLP
 - NLP Tasks and Applications
 - NLP Terminology
 - Word Embeddings
 - Language Modelling
- Part 2 (NLP and Digital Traces)
 - What are digital traces?
 - NLP case studies and applications

$\mathbf{u}^b$ Part 1 (Introduction to NLP)

Part 1 (Introduction to NLP)

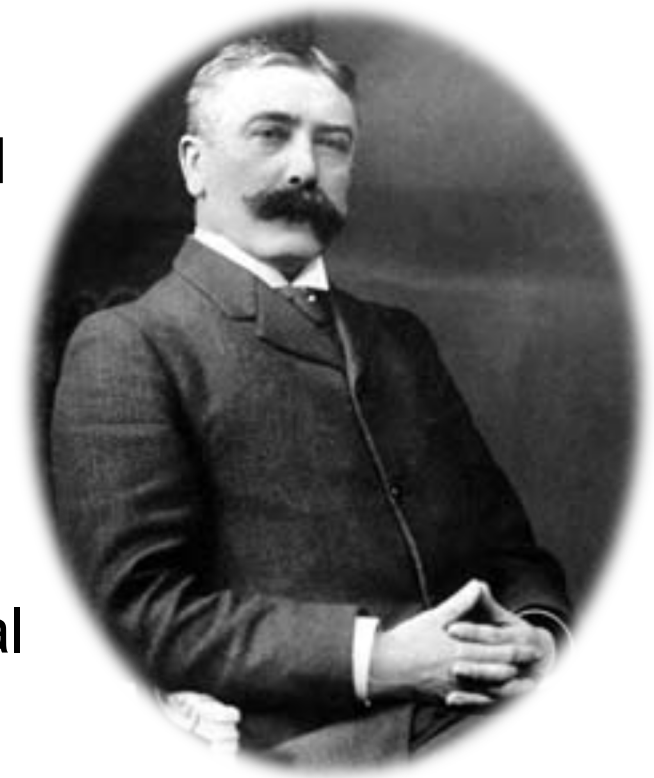


u^b What is a Language?

u^b What is a Language?

Language

- is a distinct, social system which functions as a shared agreement among members of a community.
- connects auditory images (signifier) to concepts (signified) and must be learned through gradual apprenticeship.
- is homogeneous, comprising a system of psychological signs uniting meanings and sound-images.
- is concrete, existing in the brain and capable of being represented in tangible forms such as writing.



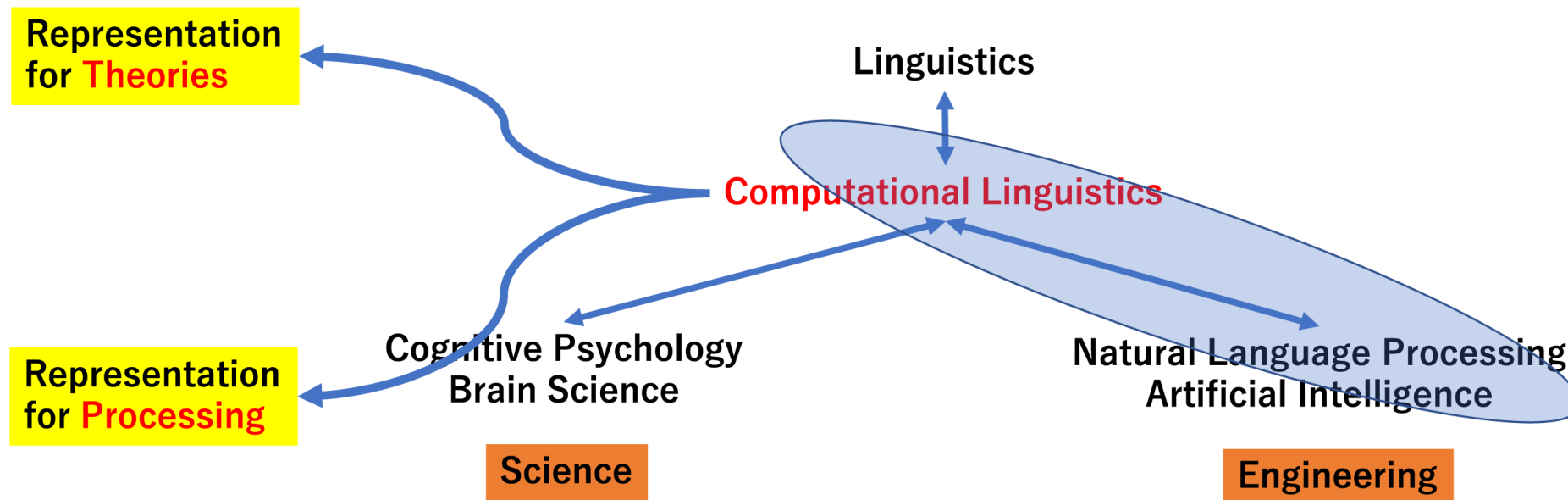
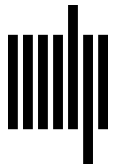
On Word Formation

Around 1900, in New Berlin, Ohio, a department-store worker named J. Murray Spangler invented a device which he called an *electric suction sweeper*. This device eventually became very popular and could have been known as a *spangler*. People could have been *spanglering* their floors or they might even have *spanglered* their rugs and curtains. The use could have extended to a type of person who droned on and on (and really sucked), described as *spanglerish*, or to a whole style of behavior called *spanglerism*. However, none of that happened. Instead, Mr. Spangler sold his new invention to a local businessman called William H. Hoover, whose Hoover Suction Sweeper Company produced the first machine called a '*Hoover*'. Not only did the word *hoover* (without a capital letter) become as familiar as *vacuum cleaner* all over the world, but in Britain, people still talk about *hoovering* (and not *spanglering*) their carpets.

u^b Properties of a language

On defining properties of a language, Chomsky notes that

“The most elementary property of language is that it involves a discrete infinity meaning that there is an infinite range of possible constructions but there is no longest sentence. However, there exists a finite system of rules which operate in some fashion to characterize the unbounded range of possible expressions. These rules operate on phrases or hierarchic structures of phrases recursively.”

**Figure Legend:**

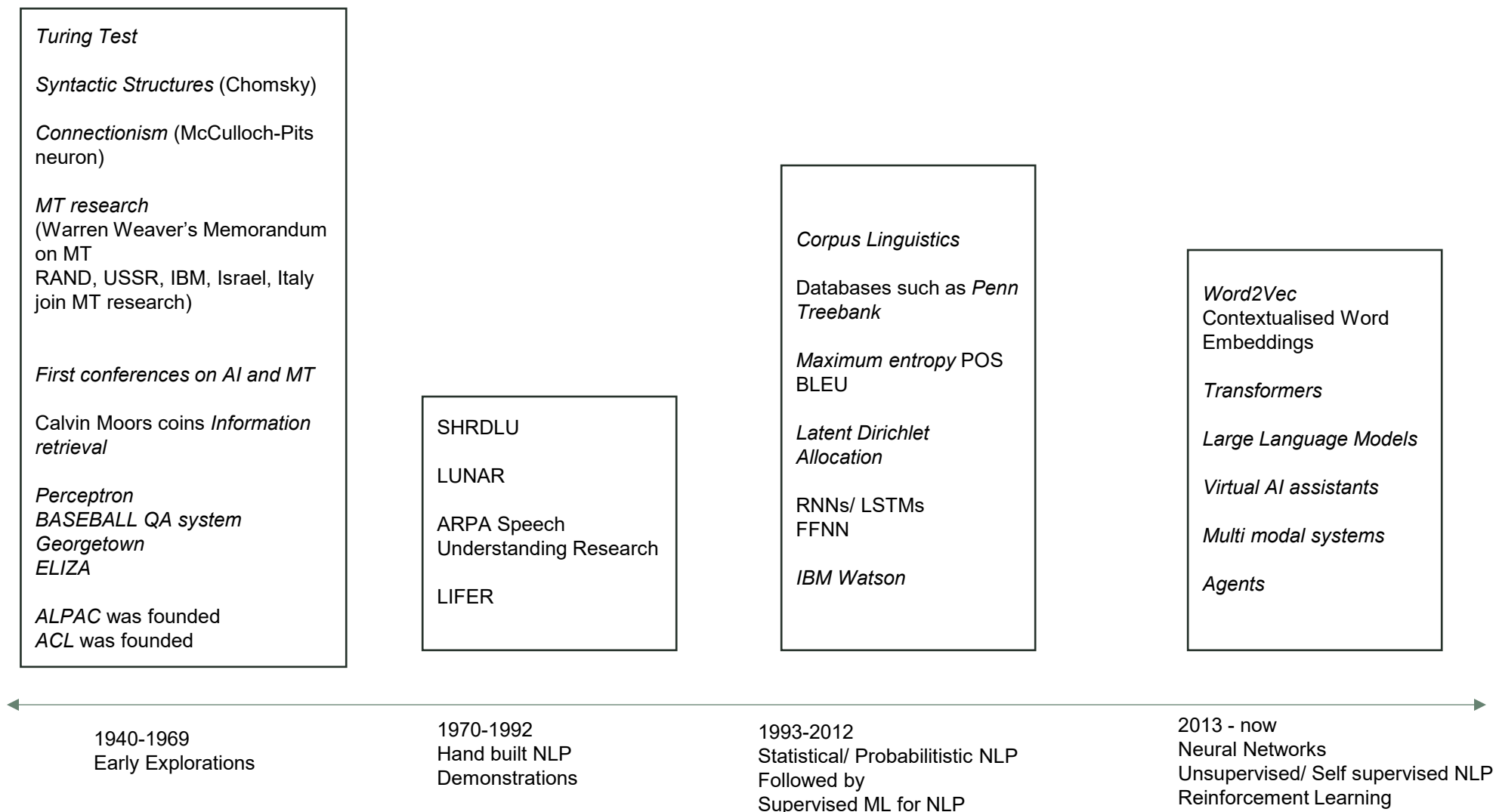
Language-related disciplines.

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What is Natural Language Processing?

Computational linguistics, also known as natural language processing (NLP), is the subfield of computer science concerned with using computational techniques to learn, understand, and produce human language content.

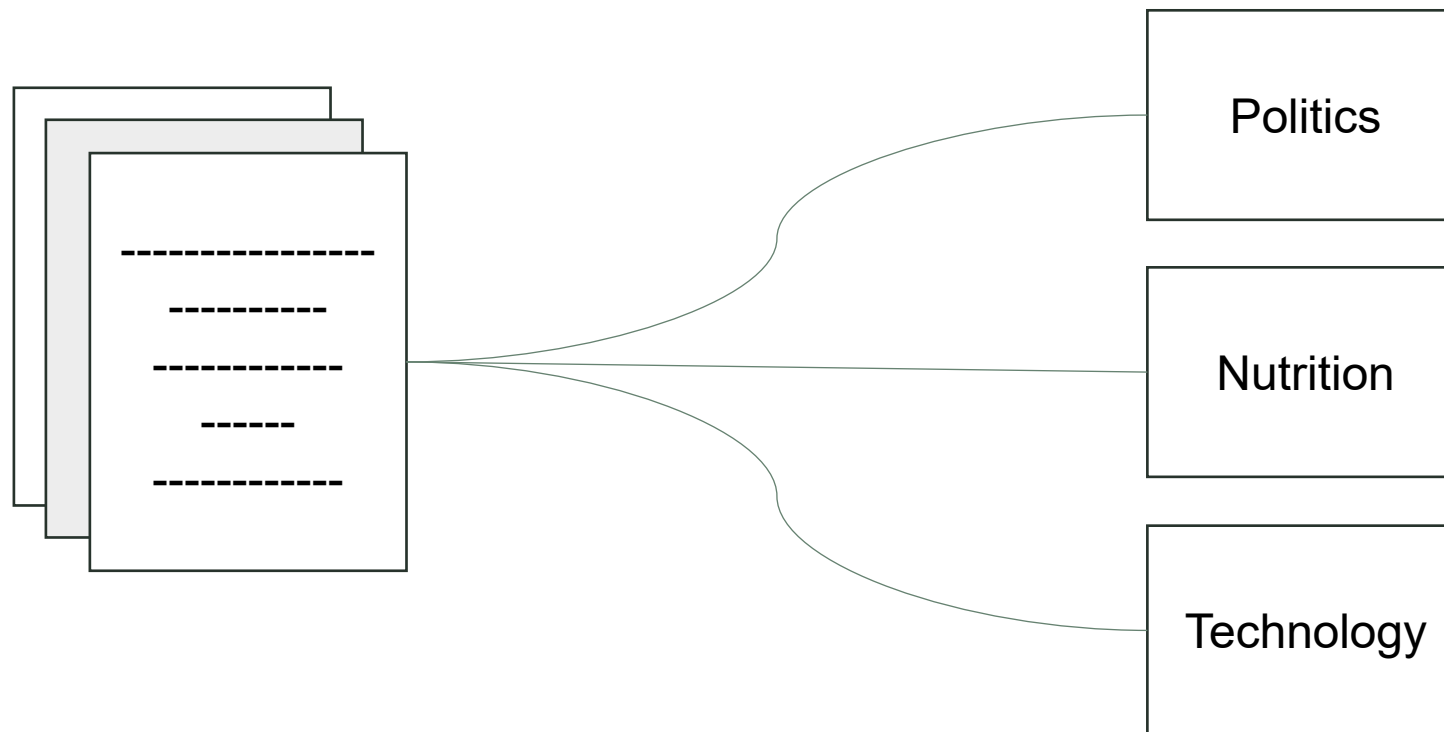
NLP Timeline



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Common NLP Tasks

Text Classification



Common NLP Tasks

Sentiment Analysis

Movie Review

All I can say is wow. I bought tickets for my wife and I because she is a huge theater fan. I do enjoy the theater myself but not overly enthusiastic. The world-building in this film is the best since Harry Potter and delivers everything you'd want from a fantasy experience.

Positive

Negative

Neutral

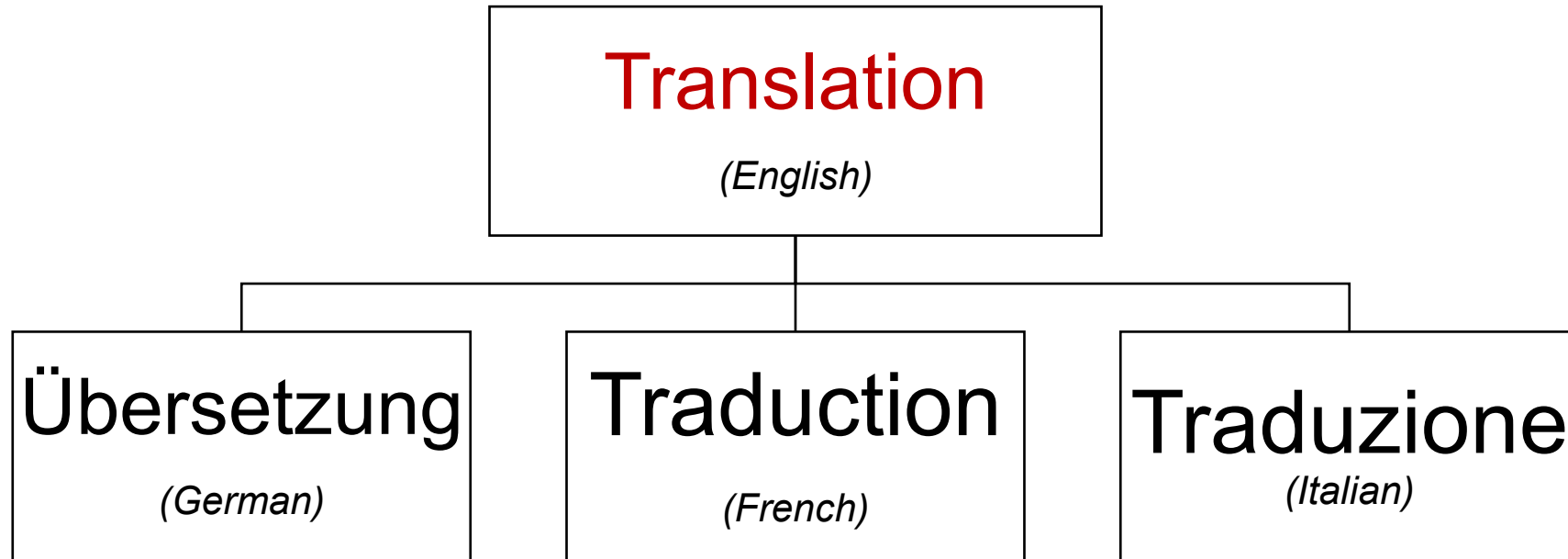
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Common NLP Tasks

Token Classification/ Named Entity Recognition

Apple **ORG** is looking at buying U.K. **GPE** startup for \$1 billion **MONEY** .

Common NLP Tasks



Common NLP Tasks

Summarization

Matthew Bourne's 1995 "Swan Lake" featured male swans, challenging ballet norms. Initially controversial, it became the longest-running full-length dance in London's West End and on Broadway, winning Olivier and Tony awards. The production celebrates its 30th anniversary with a 2024/25 tour.

BBC

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'It was a shock to many': Matthew Bourne on his Swan Lake with male swans, the show that shook up the dance world

27 November 2024

Emma Jones

Share Save



Emmanuel Lafont/ Getty Images

Celebrating the 30th anniversary of his legendary stage production, Matthew Bourne tells the BBC about the show that radically changed ballet with one "big idea".

The most iconic dance costume of recent times may be a pair of white feathery breeches on permanent display at London's V&A museum. They're a tribute to choreographer Matthew Bourne's gender-flipping dance version of the ballet Swan Lake. The production, which first premiered in London in November 1995, ruffled feathers in many ways because the swans, until then female roles, were played by male dancers. It went on to become the longest running full-length dance classic in the West End and on Broadway, winning both Olivier and Tony awards.

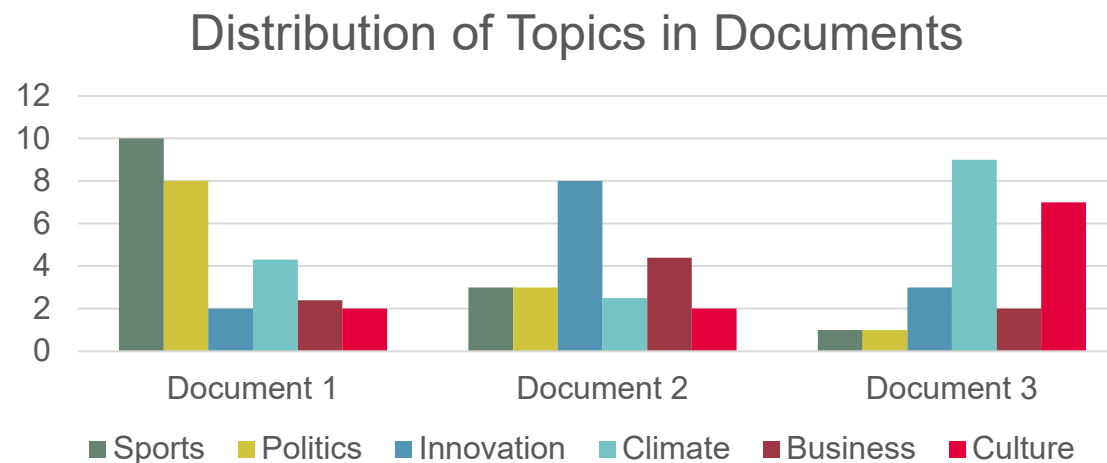
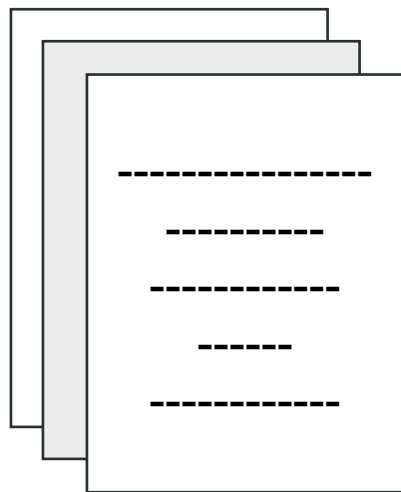
As the show celebrates its 30th anniversary with a 2024/25 tour, Matthew Bourne tells the BBC the story of the landmark production.

"I think most people thought that when they came to see it, they were going to see men in tutus," says Matthew Bourne, of public expectations back in 1995.

The young, London-born choreographer, then aged 35, had been given the opportunity to stage his own version of Swan Lake at London's Sadler's Wells Theatre. It was a cherished dream for him, he says, "because I've always loved and identified with the story". His big idea, he recalls, "was that all the swans would be male. Everything else about the production flowed from that one simple idea".

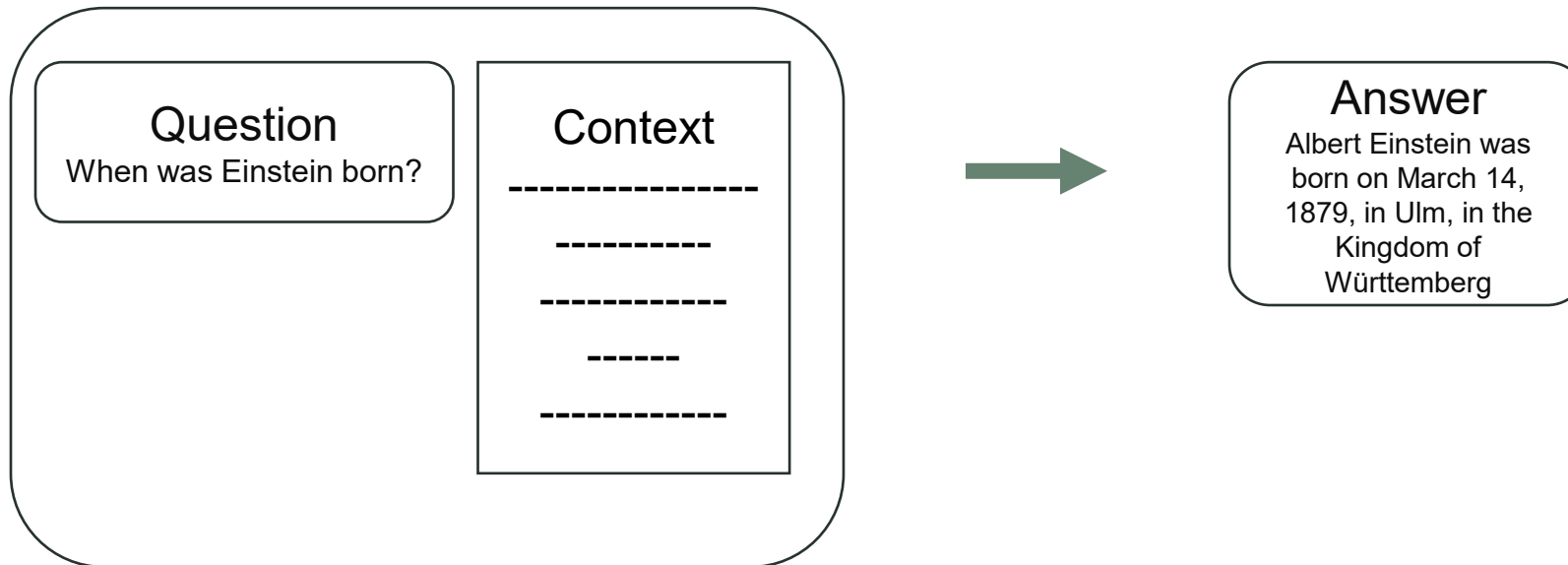
Common NLP Tasks

Topic Modelling



Common NLP Tasks

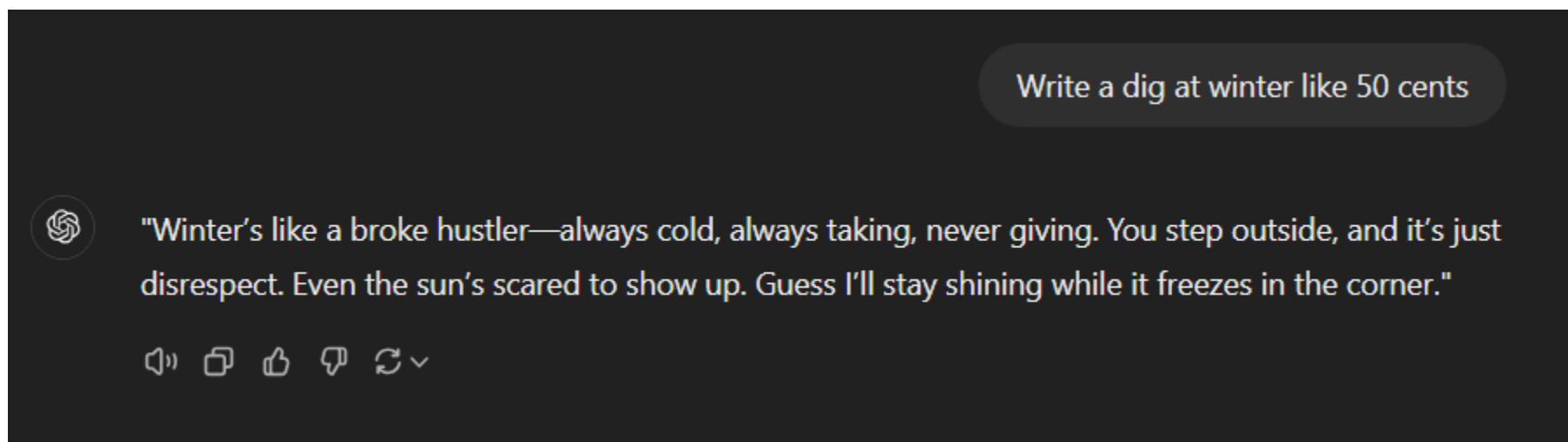
Question Answering



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Common NLP Tasks

Generation



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What are the primary challenges for NLP?

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Flying saucers can be dangerous!

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Flying saucers can be dangerous!



Larroque, P. (2023). *An introduction to linguistics through popular music*. Springer Nature.

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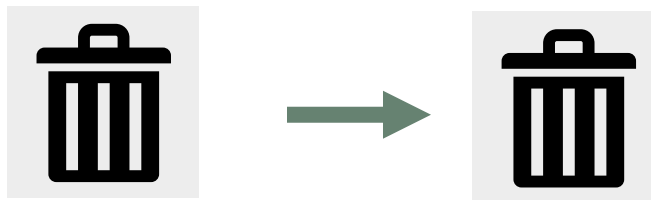
What are the primary challenges for NLP?

1. Since language is **ambiguous**, understanding language requires **context**.

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What are the primary challenges for NLP?

1. Since language is ambiguous, understanding language requires context.
2. Need of **quality** data.



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How do computers understand text?

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How do computers understand text?

They don't!

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How do computers understand text?

Rather, computers operate on a numerical representation of text represented as vectors.

Word Embeddings

A	happy	dog	is	on	the	train
122	23	841	677	2	45	63
...	...	...	...	...	...	...
11	87	1001	245	34	19	322

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Tokenization

The process of splitting text into smaller units like characters, words, sub-words, or sentences.

For example;

Sentence	Coffee is happiness.
Character	'C','o','f','f','e','e',' ','i','s',' ','h','a','p','p','i','n','e','s','s','.'
Word	'Coffee' 'is' 'happiness' '.'
Subword	'Coffee' 'is' 'happi' '#ness'

Content words, Function words and Stopwords

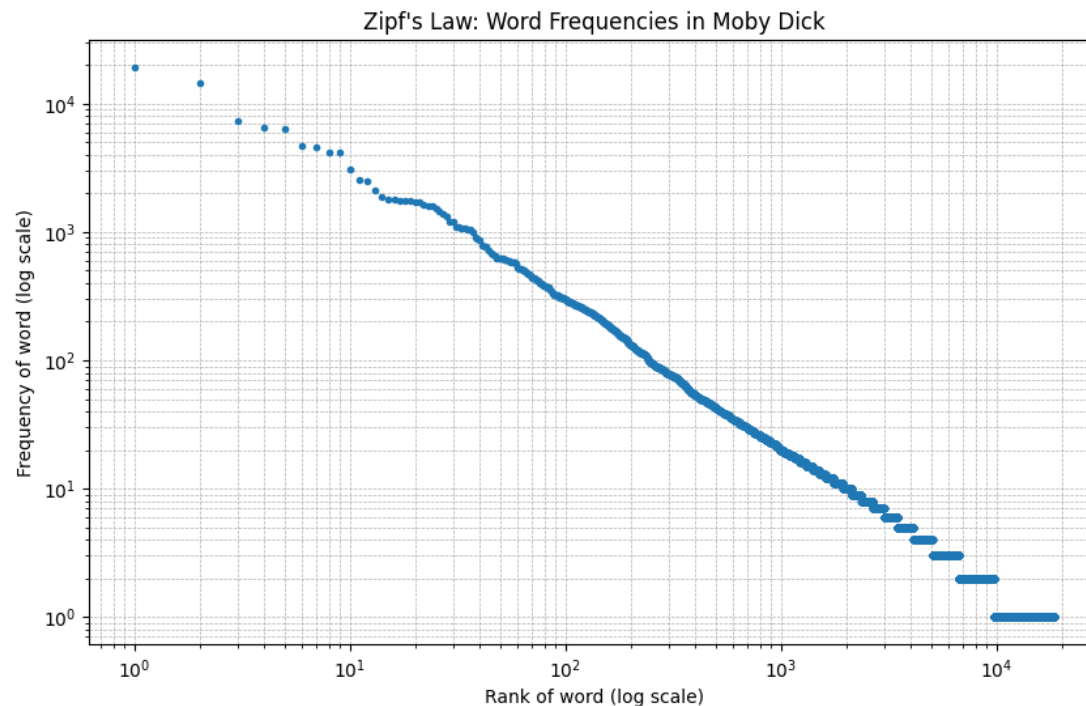
“A happy dog is on the train”

Content Words	Function Words	Stop Words
carry the semantic meaning	connect content words grammatically	are filtered out in NLP tasks due to their high frequency and low significance.
happy, dog, train	a, is, on, the	a, is, on, the

Zipf's law

$$\text{Word Frequency} = \frac{1}{\text{Word Rank}}$$

Implying that the most common word occurs twice as frequently as the second most common one and thrice as frequently as the third most common word and so on...



Stemming and Lemmatization

	Stemming	Lemmatization
Definition	reduces words to their base form by chopping off prefixes or suffixes without linguistic context.	reduces words to their root form (lemma) based on grammatical context.
a	A	a
happy	happ	happy
dog	dog	dog
is	is	be
on	on	on
the	the	the
train	train	train

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Word Embeddings

How can text be represented as numerical vectors?

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Word Embeddings

How can text be represented as numerical vectors?

Word-Level Embeddings	Contextual Embeddings	Sentence and Document-Level Embeddings	Multimodal Embeddings
These represent individual words as vectors.	These dynamically generate embeddings based on the context of the word in the sentence.	These represent sentences or entire documents.	These represent combined modalities such as text and images.
BOW TF-IDF Word2Vec Glove Fasttext	EIMo BERT	Doc2Vec SBERT	CLIP

u^b Bag-of-Words (BOW)

a	happy	dog	is	on	the	train
11	2	1	3	2	5	2

- A traditional approach where a sequence of words is represented as a vector of the frequency of occurrence of the word.
- Easy to understand and create.
- Problem of sparse vectors but reducing vocabulary means giving up on rarer but potentially information rich words.
- Cannot capture any semantic meaning.
- All words are treated equally, leading to overrepresentation of frequent but less informative words

(D1) happy dog train	(D2) cat dog	(D3) happy dog beach
-------------------------------	--------------------	-------------------------------

Given three documents with the following words, which word/words do you think are important and why?

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Term Frequency-Inverse Document Frequency(TF-IDF)

TF-IDF evaluate the importance of a token in a document relative to a collection of documents (corpus). It's a product of two components:

- Term Frequency (TF): The frequency of a word in a specific document.

$$TF(t, d) = \frac{\text{frequency count of } t \text{ in } d}{\text{total number of words in } d}$$

- Inverse Document Frequency (IDF): A measure of how rare or unique the word is across the corpus.

$$IDF(t, D) = \log\left(\frac{\text{total number of documents}}{\text{number of documents containing } t}\right)$$

Where t is a term, d is a document and D is the document collection.

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TF-IDF Example

Term	TF (D1)	TF(D2)	TF(D3)
happy	1/3	0	1/3
dog	1/3	1/2	1/3
train	1/3	0	0
cat	0	1/2	0
beach	0	0	1/3

Term	IDF
happy	$\log(3/2)$
dog	$\log(3/3)$
train	$\log(3/1)$
cat	$\log(3/1)$
beach	$\log(3/1)$

(D1)
happy
dog
train

(D2)
cat
dog

(D3)
happy
dog
beach

Term	TF-IDF(D1)	TF-IDF(D2)	TF-IDF(D3)
happy			
dog			
train			
cat			
beach			

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TF-IDF Example

Term	TF (D1)	TF(D2)	TF(D3)
happy	0.33	0	0.33
dog	0.33	0.5	0.33
train	0.33	0	0
cat	0	0.5	0
beach	0	0	0.33

Term	IDF
happy	0.18
dog	0
train	0.47
cat	0.47
beach	0.47

(D1)
happy
dog
train

(D2)
cat
dog

(D3)
happy
dog
beach

Term	TF-IDF(D1)	TF-IDF(D2)	TF-IDF(D3)
happy	0.33*0.18	0	0.33*0.18
dog	0	0	0
train	0.33*0.47	0	0
cat	0	0.5*0.47	0
beach	0	0	0.33*0.47

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TF-IDF Example

Term	TF (D1)	TF(D2)	TF(D3)
happy	0.33	0	0.33
dog	0.33	0.5	0.33
train	0.33	0	0
cat	0	0.5	0
beach	0	0	0.33

Term	IDF
happy	0.18
dog	0
train	0.47
cat	0.47
beach	0.47

(D1)
happy
dog
train

(D2)
cat
dog

(D3)
happy
dog
beach

Term	TF-IDF(D1)	TF-IDF(D2)	TF-IDF(D3)
happy	0.06	0	0.06
dog	0	0	0
train	0.16	0	0
cat	0	0.235	0
beach	0	0	0.16

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One Hot Encoding

1. Sequence of tokens can also be encoded in a binary fashion over the length of the vocabulary.
2. Easy to understand and generate.
3. However, vectors are long and sparse.
4. No semantic information.

Text	happy	dog	train	cat	beach
happy dog train	1	1	1	0	0
cat dog	0	1	0	1	0
happy dog beach	1	1	0	0	1

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Word2Vec

1. Represents words as dense, continuous vectors in a low dimensional vector space.
2. Captures contextual meanings of words based on co-occurrence with other words.
3. Captures semantic relationships between words.

$$\text{vec}(\text{king}) - \text{vec}(\text{man}) + \text{vec}(\text{woman}) \approx \text{vec}(\text{queen})$$

Dim	a	happy	dog	is	on	the	train
1	0.55	0.89	0.31	-0.21	0.65	0.90	.23
2	-0.23	0.45	-0.23	0.81	0.32	0.21	0.54
...	...	...	...	...	...	...	...
50	-0.11	0.65	0.12	0.78	0.94	0.73	-0.32

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Word2Vec Architecture

Continuous Bag-Of-Words

1. Represents words as dense, continuous vectors in a low dimensional vector space.
2. Captures contextual meanings of words based on co-occurrence with other words.
3. Captures semantic relationships between words.

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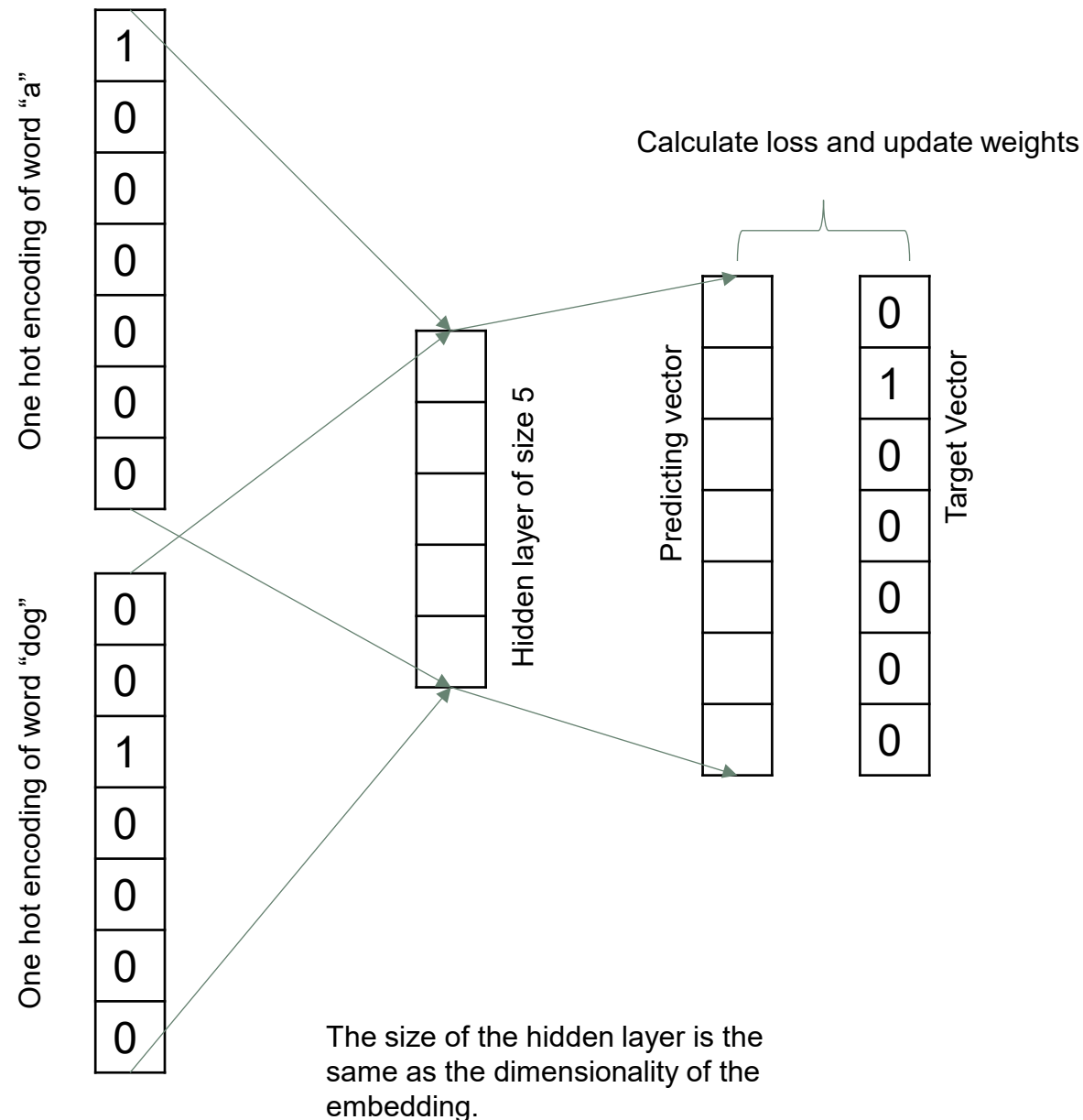
CBOW

Predicts the target word given the surrounding context

For example, in the following sentence, given a window size of 3, the goal is to predict the word “happy” when the surrounding words “a” and “dog” are known.

“a happy dog is on the train”

The window is sliding through the sequence.



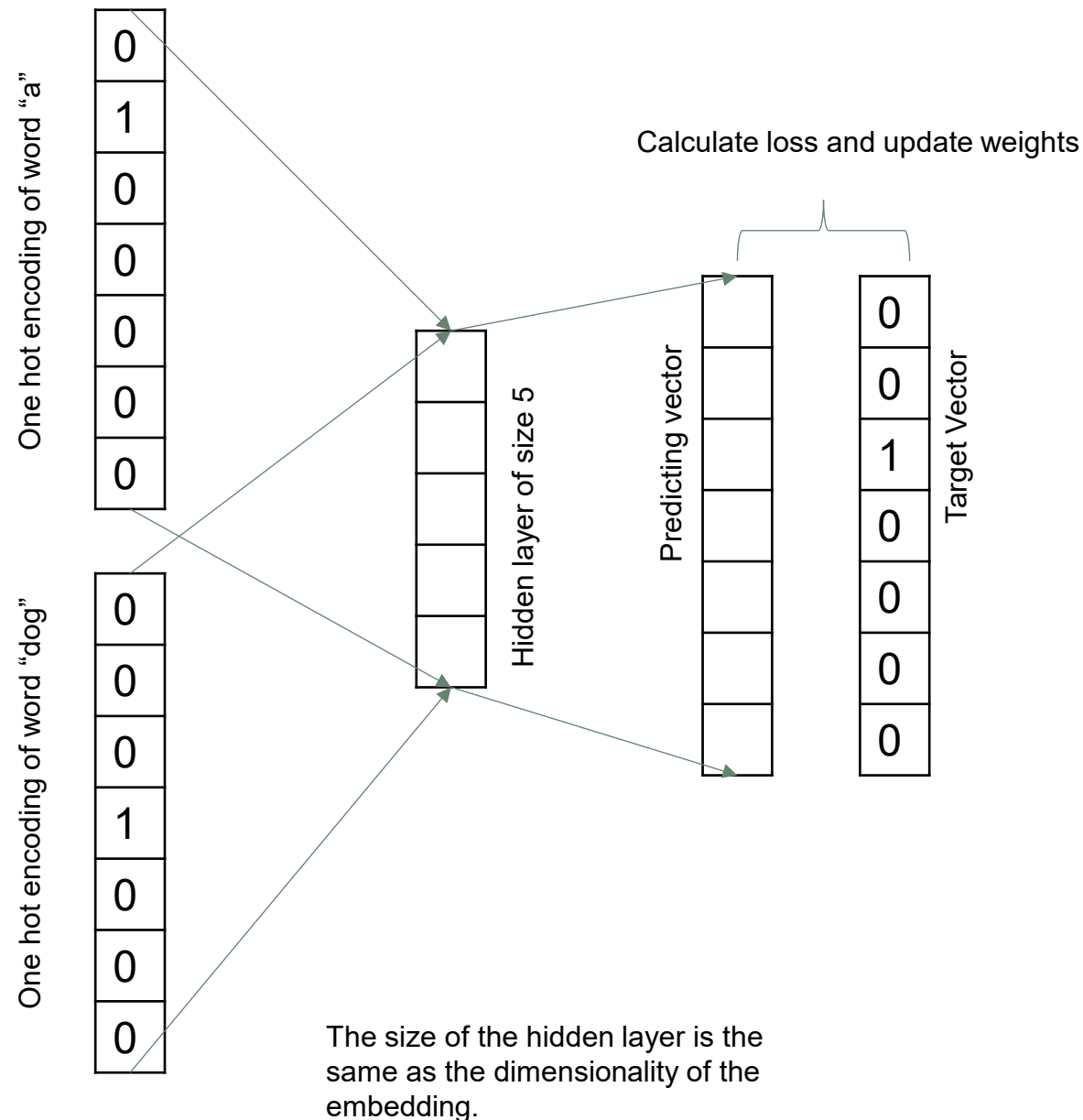
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CBOW

Predicts the target word given the surrounding context

For example, in the following sentence, given a window size of 3, the goal is to predict the word “dog” when the surrounding words “happy” and “is” are known.

“a happy dog is on the train”



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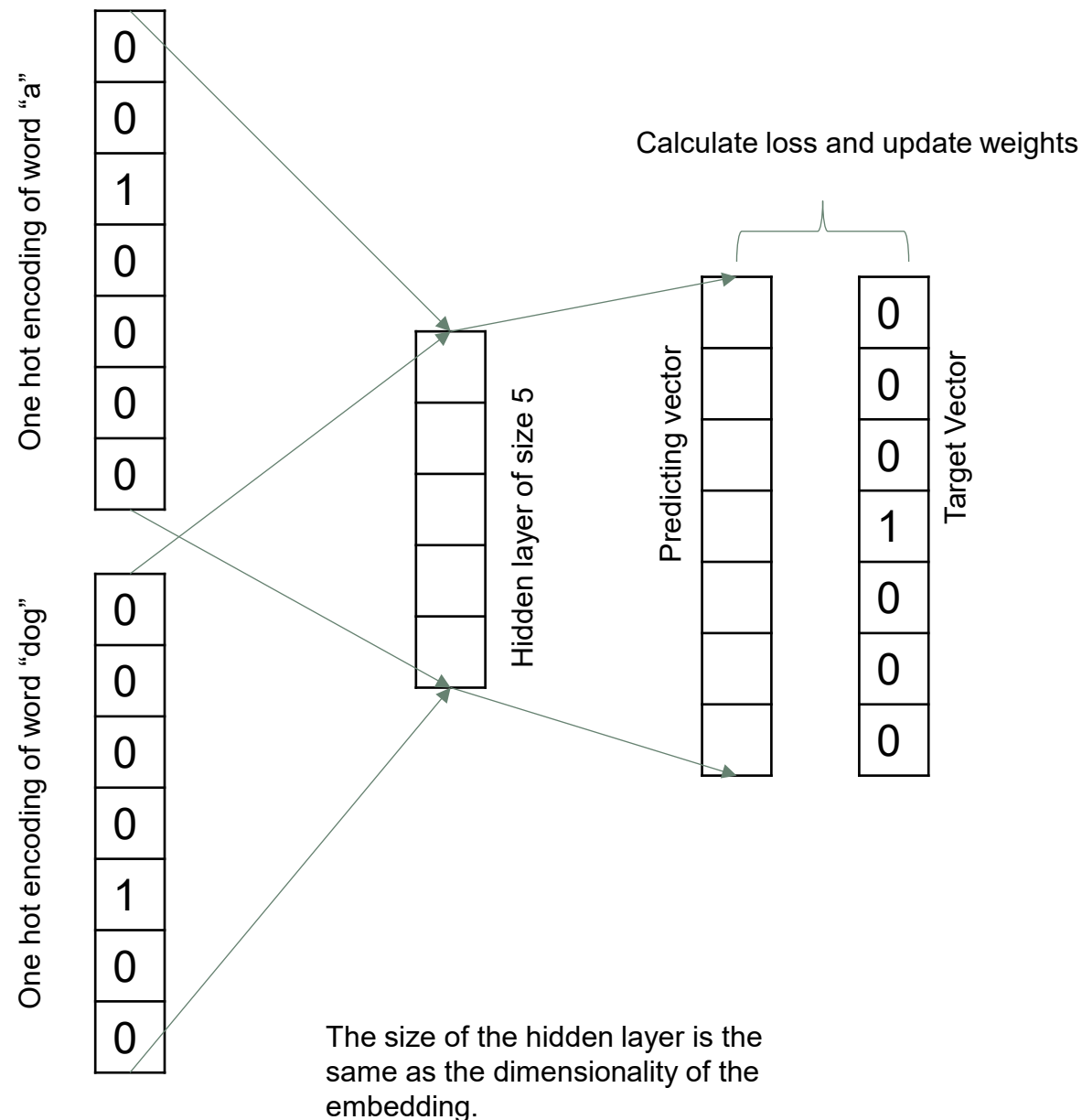
CBOW

Predicts the target word given the surrounding context

For example, in the following sentence, given a window size of 3, the goal is to predict the word “is” when the surrounding words “dog” and “on” are known.

“a happy dog is on the train”

And so on...



u^b

CBOW

Advantages of CBOW

- Works well for frequent words.
- Faster to train.



a	happy	dog	is	on	the	Train
0.11	0.29	0.66	0.23	0.69	0.71	0.10
0.28	0.71	0.79	0.81	-0.31	-0.89	0.43
0.19	0.13	0.72	-0.21	0.12	0.87	0.91
0.91	0.76	0.41	-0.31	0.54	0.52	0.71
-0.37	0.39	0.91	0.89	0.12	0.19	0.85

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Skip gram

Predicts the surrounding context given a target word.

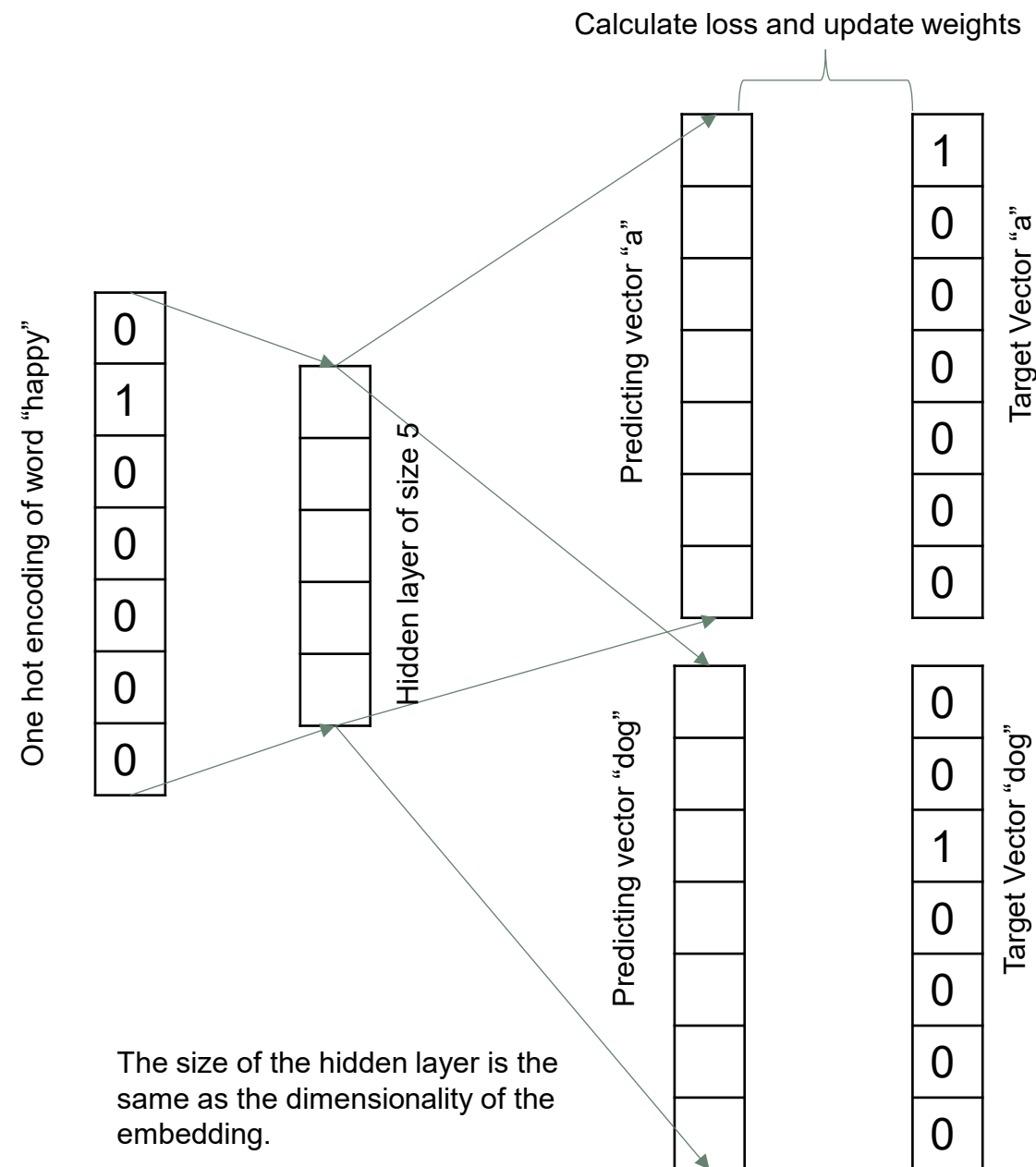
For example, in the following sentence, given a window size of 3, the goal is to predict the surrounding words “a” and “dog” when the target word “happy” is known.

“a happy dog is on the train”

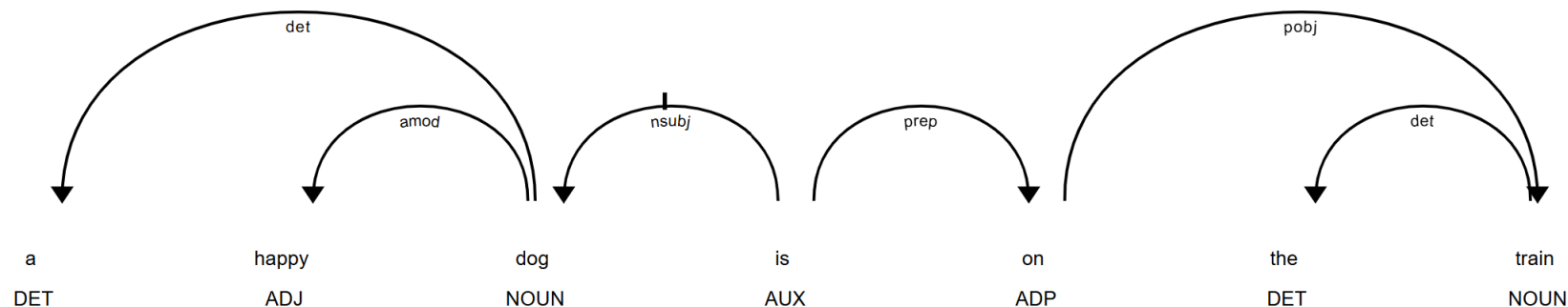
The window is sliding through the sequence.

Advantages and disadvantages:

- Works well for rare words.
- Slower to train.



Parts of Speech (POS) Tagging



POS) Tagging is the process of assigning grammatical labels (e.g., noun, verb, adjective) to each word in a sentence based on its role in the context.

POS	Explanation
DET	Determiner
ADJ	Adjective
NOUN	Noun
AUX	Auxiliary
ADP	Adposition
DET	Determiner
NOUN	Noun

Ngrams

	Word Ngrams	Character Ngrams
Sentence 'a happy dog is on the train'	N-word sequences from the tokenized sentence.	N-character sequences from the raw text.
Unigram	['a', 'happy', 'dog', 'is', 'on', 'the', 'train']	['a', ' ', 'h', 'a', 'p', 'p', 'y', ' ', 'd', 'o', 'g', ' ', 'i', 's', ' ', 'o', 'n', ' ', 't', 'h', 'e', ' ', 't', 'r', 'a', 'i', 'n',]
Bigram	Bigram words ['a happy', 'happy dog', 'dog is', 'is on', 'on the', 'the train']	['a ', ' h', 'ha', 'ap', 'pp', 'py', 'y ', ' d', 'do', 'og', 'g ', ' i', 'is', 's ', ' o', 'on', 'n ', ' t', 'th', 'he', 'e ', ' t', 'tr', 'ra', 'ai', 'in']

u^b Regular Expressions

- **Regex** are sequences of characters defining search patterns used for matching, extracting, or replacing text within strings. They are widely used in text processing, data validation, and parsing.
- Literal Characters match exactly as written .Example: cat matches "cat" in "catalog."
- Metacharacters are special characters with specific meanings: Some examples are:
 - '.' Matches any character except a newline.
 - '*' Matches 0 or more occurrences of the preceding element.
 - '+' Matches 1 or more occurrences of the preceding element.
 - '?' Matches 0 or 1 occurrence.
 - '[' Matches any single character in the brackets.^: Matches the start of a string.\$: Matches the end of a string.Escaping: Use \ to escape special characters (e.g., \. matches a literal period)

<https://regex101.com/>

<https://thinkwithgames.itch.io/regex-adventure-browser>

[re — Regular expression operations — Python 3.13.1 documentation](#)

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Language Models

Language models estimate the likelihood of the next word given previous words.

$$P(\textit{The}, \textit{sea}, \textit{is}, \textit{red}) = 0.8$$

$$P(\textit{The}, \textit{red}, \textit{is}, \textit{sea}) = 0.2$$

$$P(\textit{sea}, \textit{is}, \textit{The}, \textit{red}) = 0.05$$



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Language Models

The joint distribution of the sequence of words can be calculated mathematically with the chain rule of probability.

For example,

$$\begin{aligned} P(The, sea, is, red) &= P(The)P(sea|The)P(is|The\ sea)P(red|The\ sea\ is) \\ &\approx P(The)P(sea|The)P(is|sea)P(red|is)^* \end{aligned}$$

*Approximation based on Markov's assumptions

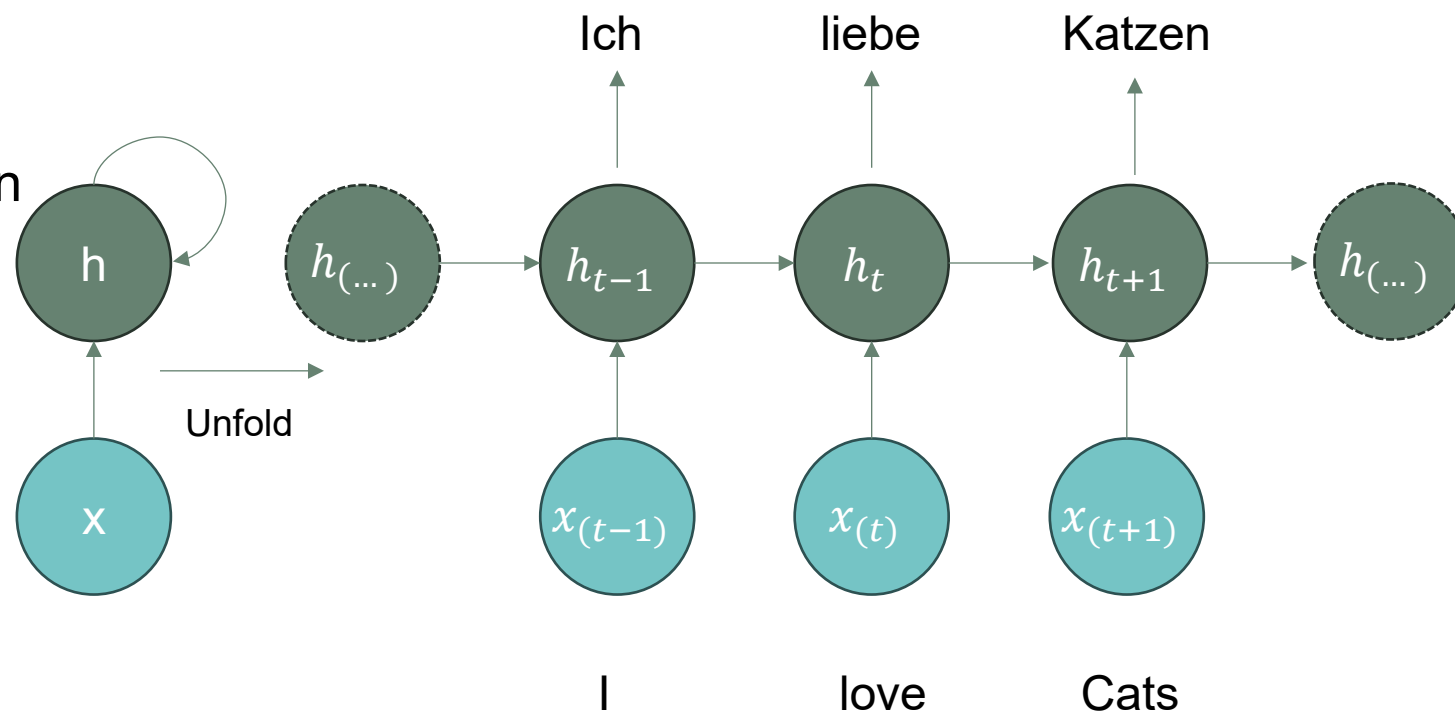
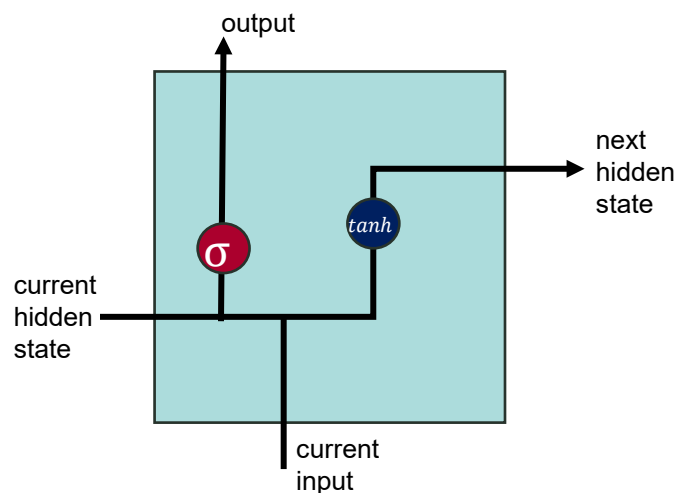
1. <https://web.stanford.edu/~jurafsky/slp3/3.pdf>

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Recurrent Neural Networks (RNNs)

RNNs introduce a latent variable called a hidden state, such that

- the output is a function of the current hidden state.
- the current hidden state is a function of the previous hidden state and the current input.

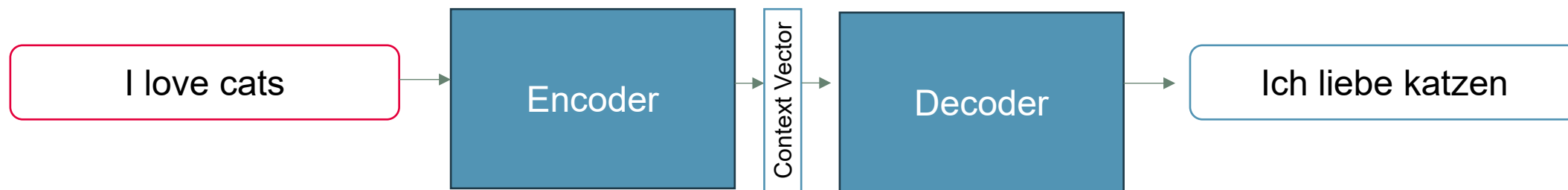


u^b Shortcomings of RNNs

1. Vanishing or exploding gradients during backpropagation through time.
2. Difficulty in accessing information from a long time ago.
3. Sequential processing word by word slows down training and inference as computation for time step t cannot be done until the computation for the time step $t-1$ has been completed.

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Sequence to Sequence Models

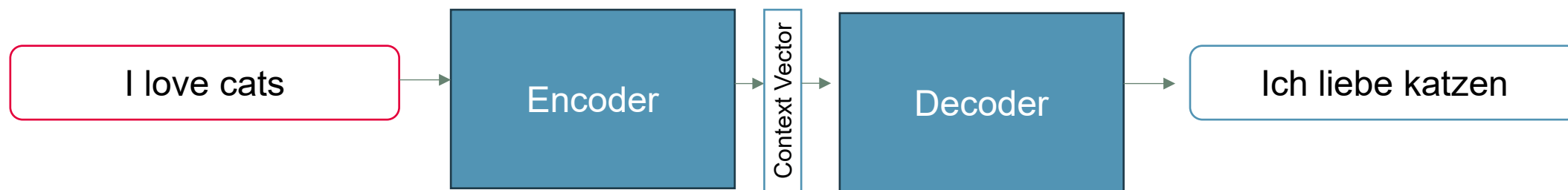


Separating encoding from decoding has the following advantages

1. The decoder now has access to the full context before beginning decoding.
2. Parallel encoding and decoding is now possible for different sequences.

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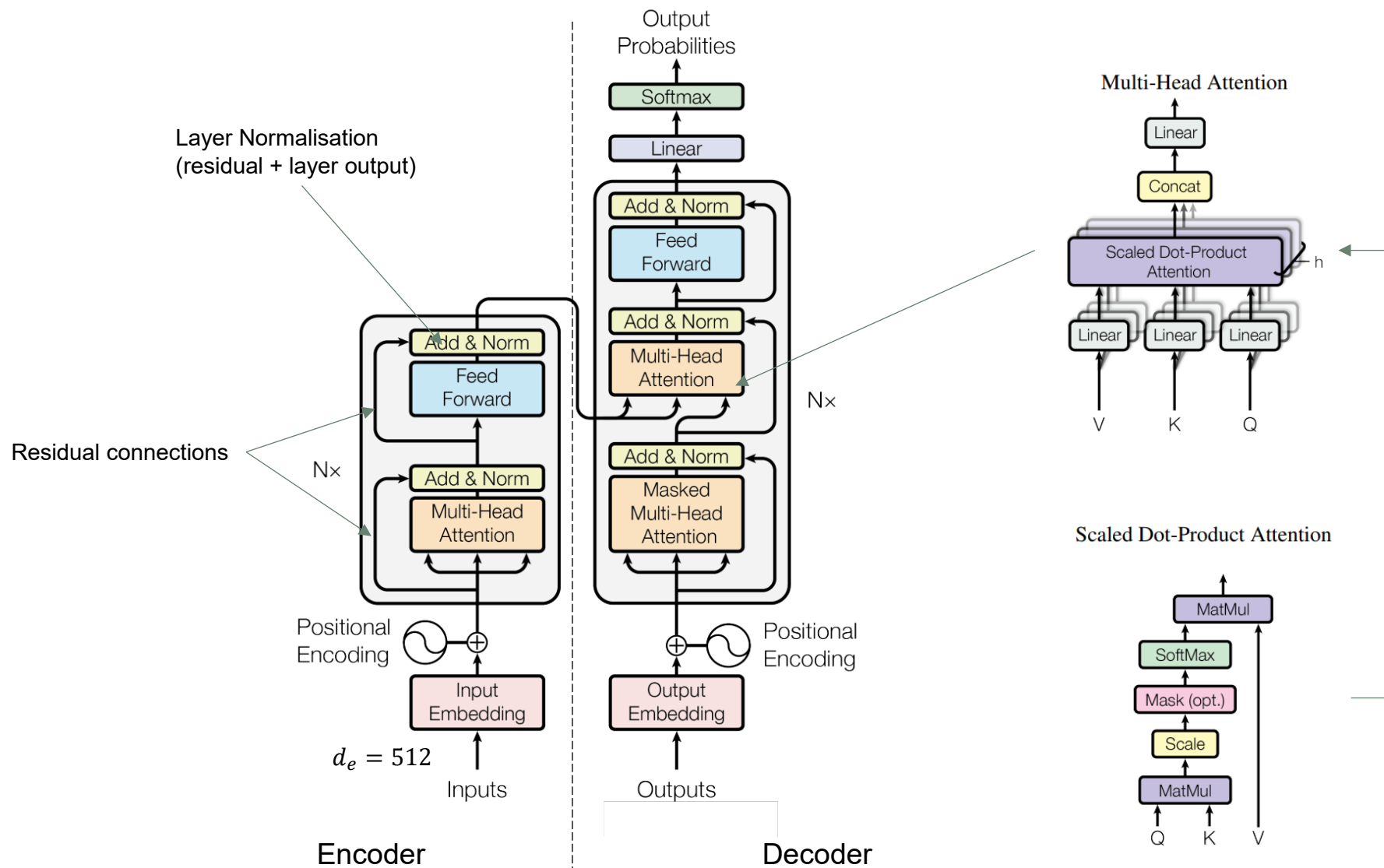
Sequence to Sequence Models



However , now there is a higher complexity....where should we pay **attention**?

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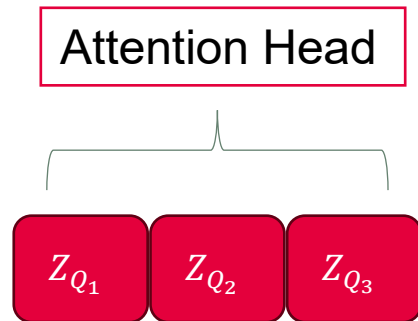
Transformer Model



Picture Credits: Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). Attention is all you need.

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Self Attention



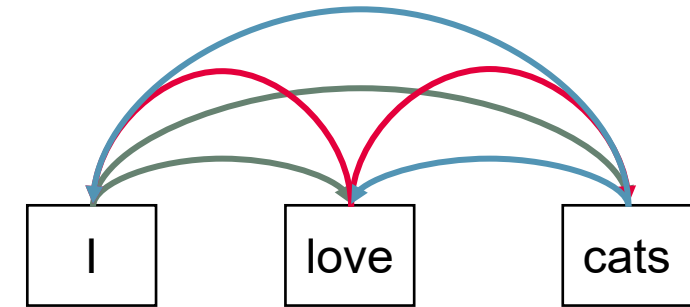
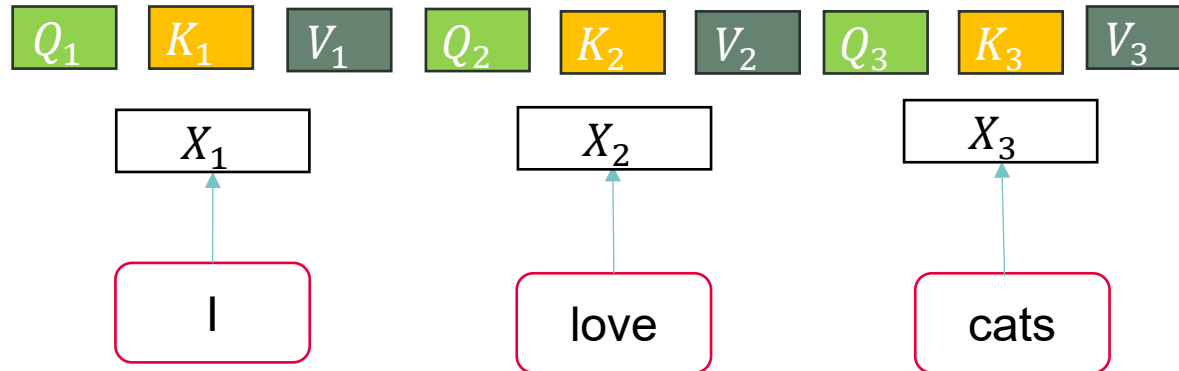
$$\text{Attention Matrix } A(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

where $Q_i = W_q X_i$ (Query),

$K_i = W_k X_i$ (Key),

$V_i = W_v X_i$ (Value),

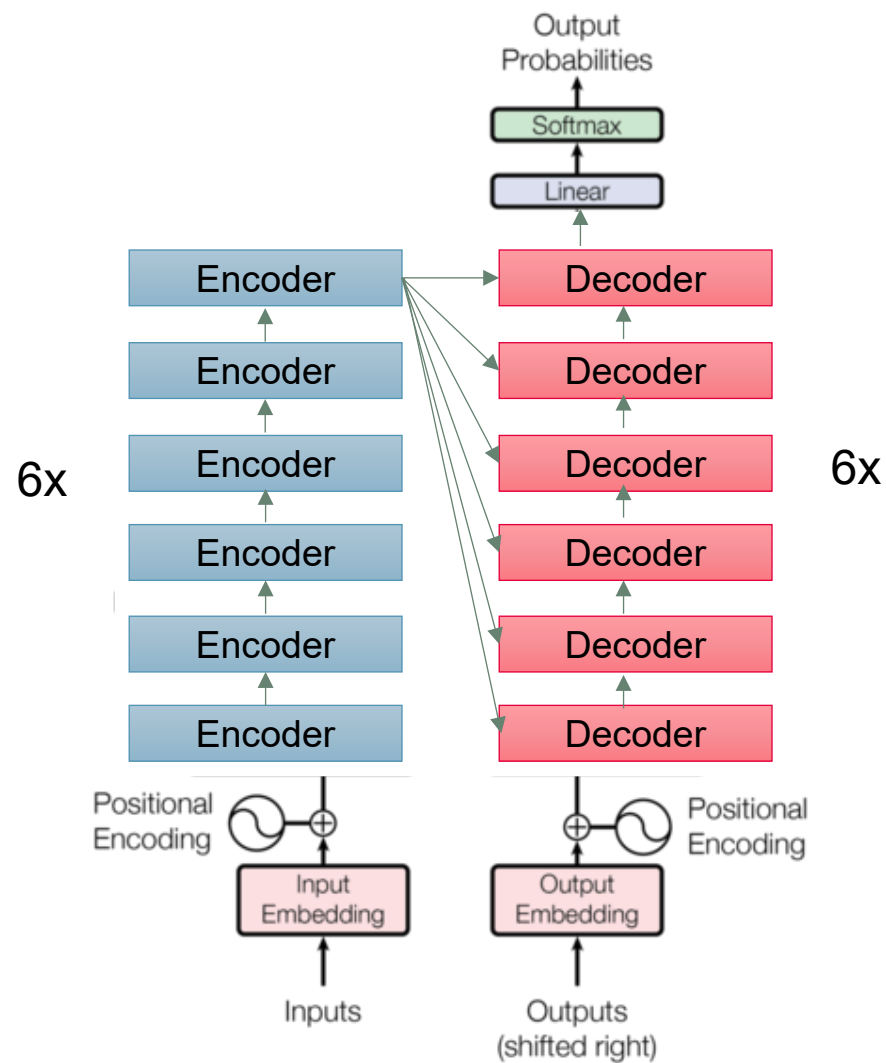
$\sqrt{d_k}$ is the square root of the key dimension
 W_q, W_k, W_v are weight matrices.



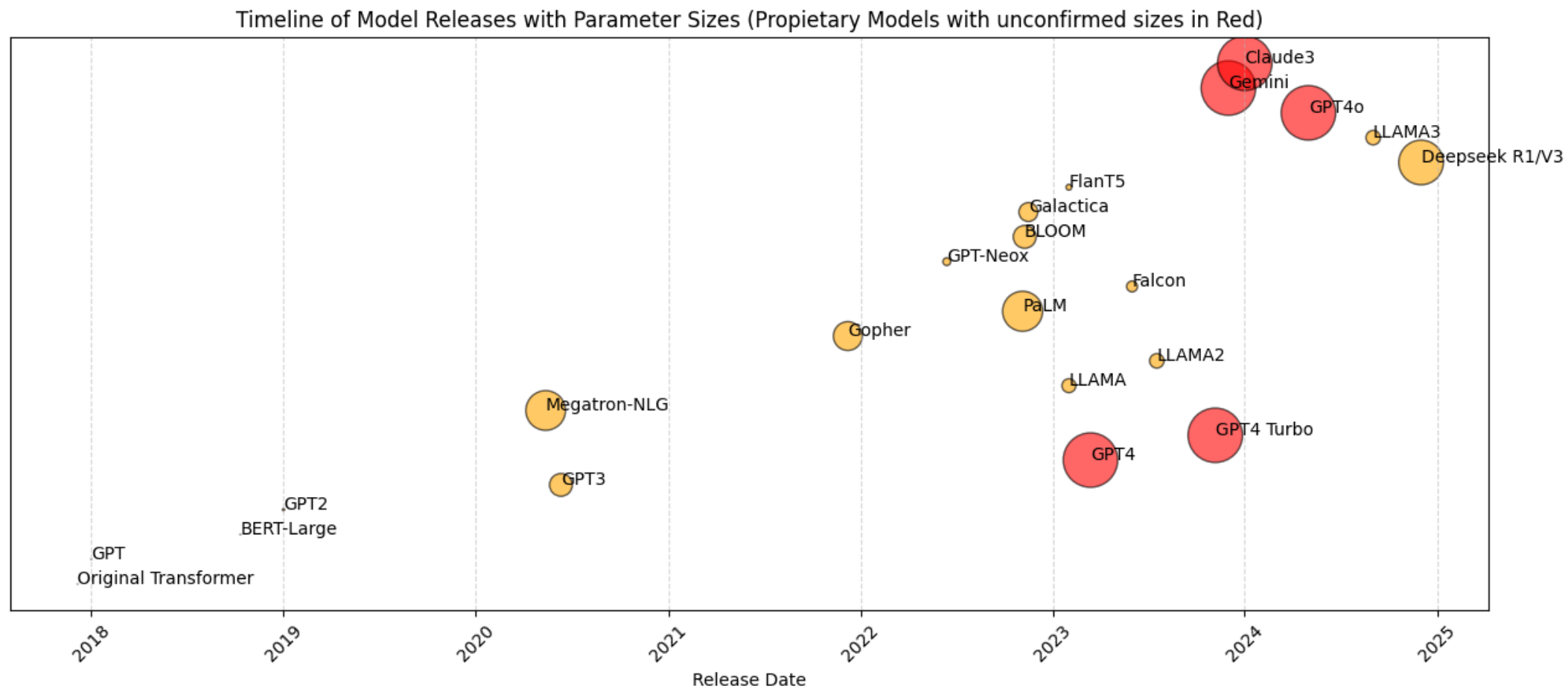
Encoder Self-Attention

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Encode and Decoder Stacks



LLM Timeline



LLM Timeline

Model Name	Estimated Release Date	Number of parameters
Original Transformer	06/12/2017	65M
GPT	01/01/2018	100M
BERT-Large	11/10/2018	110M
GPT2	01/01/2019	1.5B
GPT3	11/06/2020	175B
GPT4*	14/03/2023	1T+?
GPT4 Turbo*	06/11/2023	1T+?
Megatron-NLG	13/05/2020	530B
LLAMA	01/02/2023	65B
LLAMA2	18/07/2023	70B
Gopher	08/12/2021	28B
PaLM	05/11/2022	540B
Falcon	01/06/2023	40B
GPT-Neox	14/06/2022	20B
BLOOM	09/11/2022	176B
Galactica	16/11/2022	120B
FlanT5	01/02/2023	11B
Deepseek R1/V3	01/12/2024	671B
LLAMA3	01/09/2024	70B
GPT4o*	01/05/2024	1T+?
Gemini*	01/12/2023	1T+?
Claude3*	01/01/2024	1T+?

Generative AI

What are Large Language Models ?

LLM models

- Contain hundreds of billions or more parameters.
- Are trained on large amount of data.
- Show an understanding of language and are suited to solve complex tasks by following instructions and in context information.
- Are mostly decoder architecture based.
- Examples are GPT-4, PaLM, LLaMA.

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Why models continue to get larger?

Emergent abilities

1. Emergent abilities are those abilities in LLMs which appear in large models but are absent in smaller models.
2. Examples of emergent abilities
 - In context learning: Model can carry out the instructions in the prompt without additional training.
 - Zero shot, One shot and Few shot learning.
 - Instruction following: LLMs can perform well on unseen tasks.
 - Step-by-step reasoning: Chain-of-thought prompting strategy LLMs can show intermediate reasoning steps.

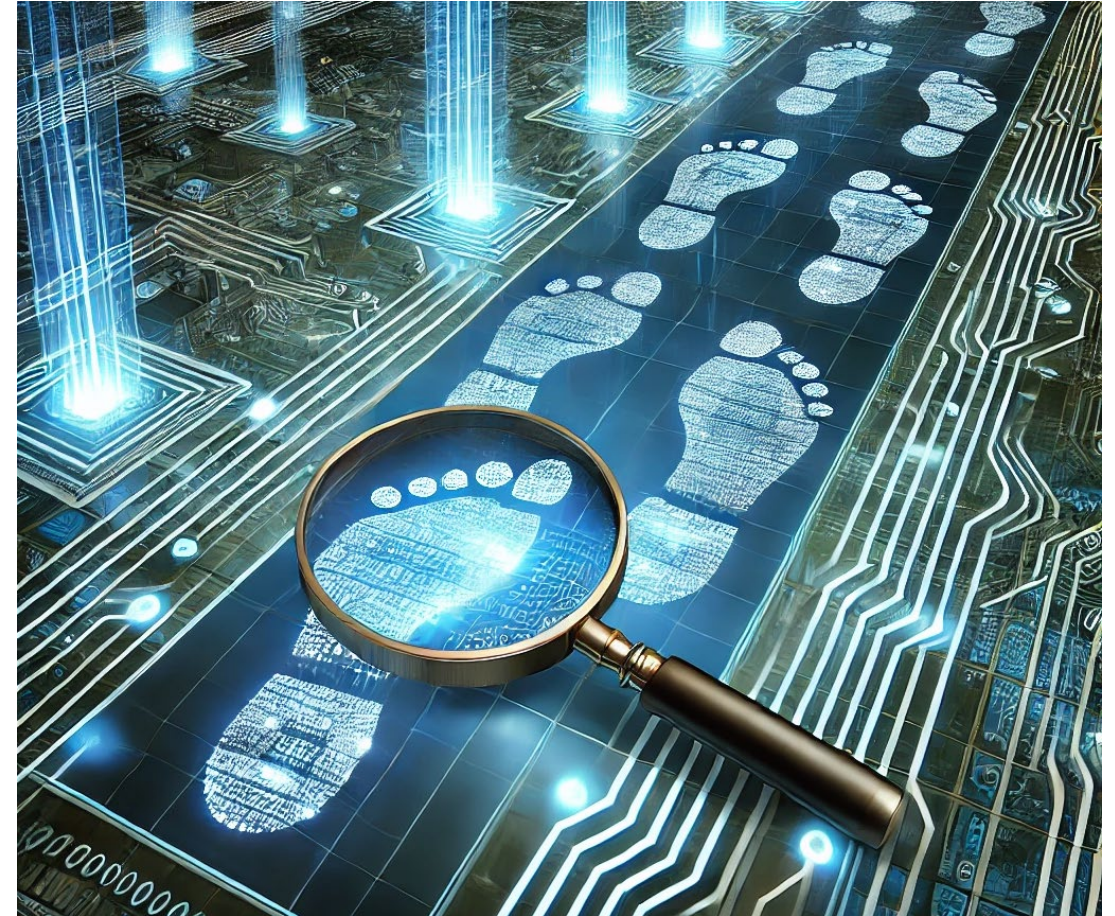
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Part 2 (NLP and Digital Traces)



What are Digital Traces?

- By interacting with the digital world, we create traces or “digital footprints” that can be used to uniquely profile or identify a person.
- Some examples of digital traces are
 - Posting and uploading pictures, on social media.
 - Online activity while using the internet or inside an app.
 - Via interactions with persons or digital entities.
- Textual digital traces are not just social media posts but also reviews, forms, survey, news articles, blogs, forums etc.
- Digital Phenotyping [1,2] is defined as the *“moment-by-moment quantification of the individual-level human phenotype in situ using data from personal digital devices”*.



[1] Torous, J., Kiang, M. V., Lorme, J., & Onnela, J. P. (2016). New tools for new research in psychiatry: a scalable and customizable platform to empower data driven smartphone research. *JMIR mental health*, 3(2), e5165.

[2] Onnela, J. P., & Rauch, S. L. (2016). Harnessing smartphone-based digital phenotyping to enhance behavioral and mental health. *Neuropsychopharmacology*, 41(7), 1691-1696.

What information can we gather from Digital Traces?

1. **Sentiment and opinions:** Text analysis can gauge overall sentiment (e.g., What is the government opinion on climate initiatives? What is the public reaction to an event?).
2. **Evolution of trends:** By monitoring keywords and topics, we can detect trends or viral discussions in society early on.
3. **Behavior patterns:** Language use might reveal patterns (e.g., increased use of anxious words during exam seasons, how do individuals claiming to have a particular health disorder express themselves in a dedicated subreddit?)

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Mental Health Detection (Case Study)

- Researchers and health professionals use NLP to find linguistic clues of mental well-being in social media.
- **Example: Depression detection on Reddit**
 - Tadesse et al. 2019 found that on reddit, *“predictors of depression contained the words related to preoccupation with themselves, feelings of sadness, anxiety, anger, hostility or suicidal thoughts, with a greater emphasis on the present and future”*.
 - NLP techniques: BOW, Ngrams, Topic Modelling, Classification

Discourse Analysis (Case Study)

Example: Oyeode et al. 2022 collected posts from multiple social media platforms during the pandemic and applied NLP to identify the negative and positive issues, opinions and perceptions. Thereafter, suitable interventions for each of the negative sub themes was given.

NLP techniques: Keyphrase extraction and categorization

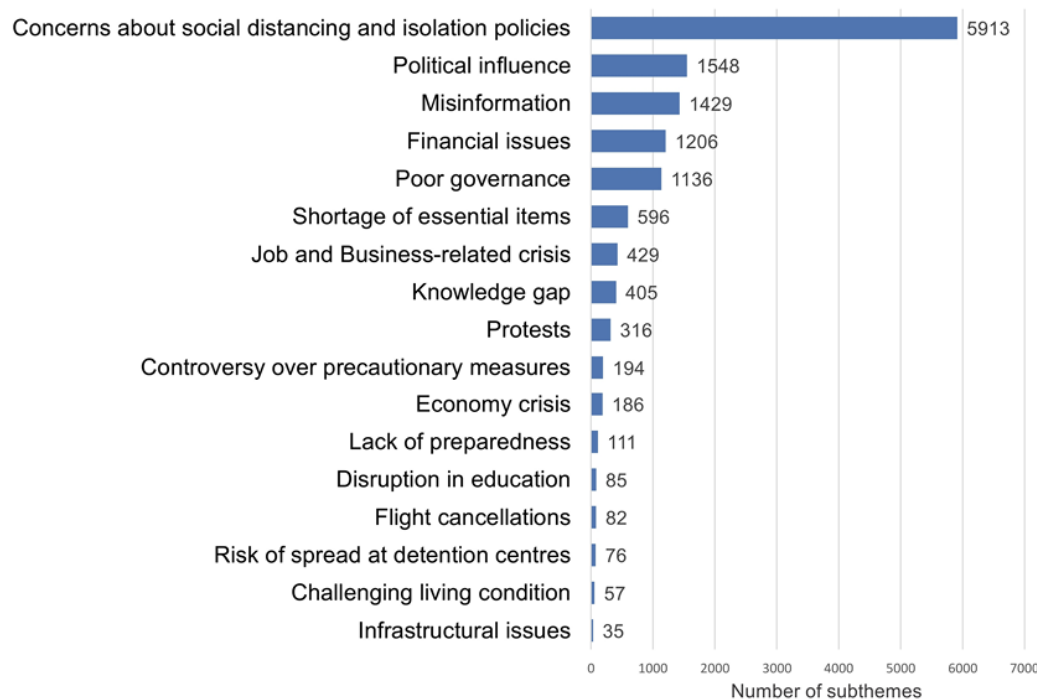


Fig. 6 The chart shows negative themes and the corresponding number of subthemes

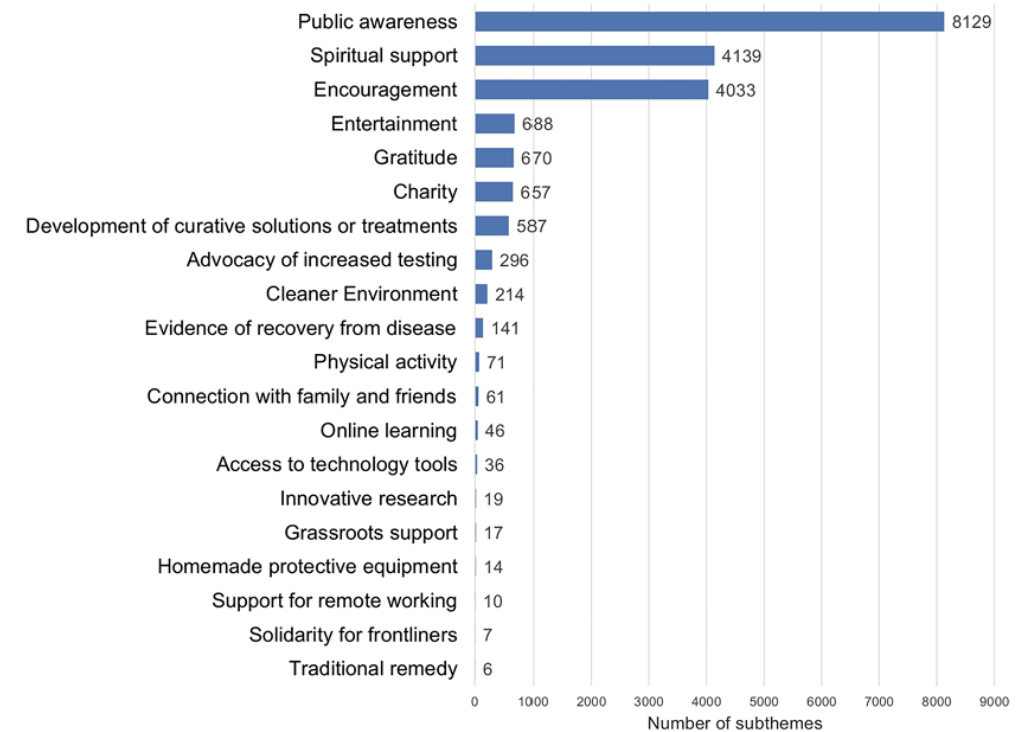


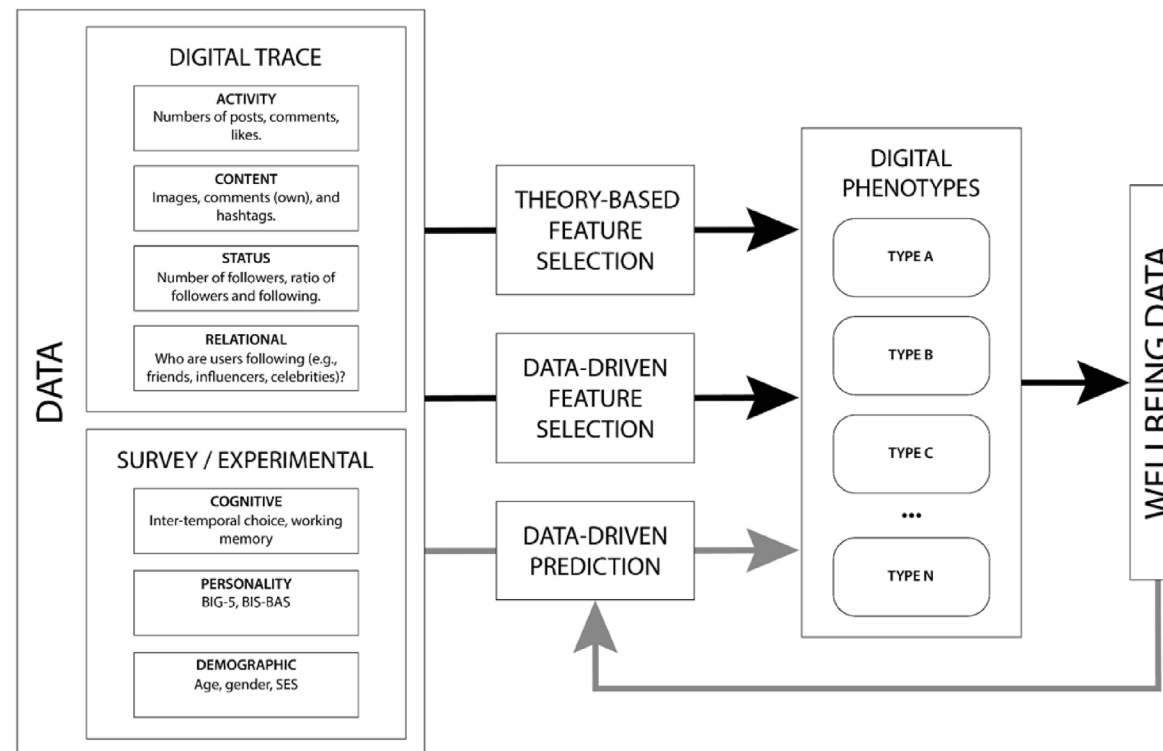
Fig. 8 The chart shows positive themes and the corresponding number of subthemes

Digital Phenotyping (Framework)

Example:

Sultan et al. 2023

- highlight the uncertainty and mixed findings on social media studies for adolescents
- propose a framework to create digital phenotypes describing types of social media users clustered by their digital trace data.



Qualitative Coding (Case Study)

Example:

- Morgan et al 2023 used ChatGPT to reinvestigate two qualitative datasets that were previously analyzed by more traditional methods.
- They found that ChatGPT performed reasonably well, but in both cases, *“it was less successful at locating subtle, interpretive themes, and more successful at reproducing concrete, descriptive themes”*

Ethics and Bias

1. Digital footprints can be considered personal especially if personal information is involved.
2. Even if posts are public, analyzing them at scale can invade perceived privacy.
3. Responsible NLP practice anonymizes data, removing personal information when analyzing behavior trends and seek compliance with ethics committee and privacy laws.
4. Social media data availability can influence that data is gathered from only certain platforms and therefore biased.

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Future Trends – How can Generative AI influence behavior analysis?

- LLMs can analyze large amounts of text (dataset of discussions or articles) and summarize, perform qualitative encoding and identify themes.
- Possibility of combining multiple data modes and embedding in the same space to generate insights.
- On the other hand, using LLMs to write can deter inferring personality traits.